

Hence, virtualisation is the major entry point for the private cloud. However, the trap that executives often fall into is when they believe virtualisation is a cloud environment. By itself, it does not constitute a private cloud.

Zynga, a provider of social game services, was founded in July 2007. It moved to the public cloud to accommodate the rapid scale demanded through the vertical growth phase of the company. In this process, it gained a better understanding of workloads in the public cloud environment and opted to build its own internal cloud to complement its public cloud footprint. The reason behind this move was a desire to recoup some of the capital expenditure made in the earlier stages of setting up the company.

Also, as the business had grown, it became feasible from a financial perspective to own some of the equipment and enjoy the associated benefits of the higher depreciation write-offs that come with it. At the same time the reduced dependence on the public cloud improved the company's operating margins. Zynga's private cloud project went from concept to production in less than six months. Zynga, along with Rightscale, a cloud management product, can now fully provision more than 1,000 physical servers in 24 hours. This is an iconic private cloud case study.

Private cloud adoption also resulted in greater control and flexibility over the storage and Web tiers, provisioning times got slashed, while shorter development cycles led to faster time to market, improved application performance and ease of management.

Private clouds and public clouds - a comparison

A comparison of the two can be made from various perspectives, which may differ based on the application type and security requirements.

- **Governance:** In the private cloud, there is better control over data (and data security); while in the public cloud, cloud service providers are responsible for many aspects related to securing infrastructure and data.
- **Speed and agility:** Both public and private clouds have the ability to balance and provision resources with very little turnaround time.
- **Risk management:** Private clouds are behind the organisation's firewall and, hence, have better risk control compared to public clouds.
- **Compliance:** Private clouds mitigate the security and privacy issues related to regulated workloads running outside the firewall by giving IT departments efficient and complete control over resources.
- **Culture:** Private clouds are better aligned to the command-and-control cultures and expectations of enterprise IT organisations. Familiar environments are easy to manage.
- **Cloud economics:** Private clouds leverage existing infrastructure investments, and this is far more cost-effective for long-running workloads. It turns out those 'pennies an hour' that you pay to public cloud service

providers add up remarkably fast, because the price you pay includes all of their costs, in addition to some healthy profit margins—though for very compute-intensive Hadoop style applications, public clouds can offer a cost advantage.

- **CapEx:** Public clouds have better resource utilisation rates; with private clouds, the organisation still has to build and maintain resources for computing, storage and networks to meet spikes in demand across various business units. There are costs associated with maintenance of resources, planning for redundancy to avoid data loss and disasters, and additional experts to handle all operations on the premises.
- **Adequate resources:** Public clouds keep infrastructure costs low for new projects while for private clouds, organisations need to obtain resources to meet requirements for unplanned projects that may crop up.
- **Elasticity:** Public clouds offer greater elasticity and provide seemingly unlimited resources for use, while this is not the case in private clouds.
- **Economies of scale:** Public clouds have greater economies of scale compared to private clouds.

The private cloud often evolves into the hybrid model that enables organisations to retain ownership of applications and data. It is the commencement of an organisation's cloud journey and not the final destination.

Now, let's take one step forward by revisiting the history of Amazon's public cloud. How did Amazon start its operations and begin offering its public cloud services? Well, it all started with the purpose of gaining flexibility and agility in the traditional environment. This is the same reason IT organisations are building private clouds today. An interesting scenario occurred in 2012, when a major public cloud provider announced its partnership with a private cloud vendor. This was an alliance between one of the biggest cloud service providers and one of the biggest private cloud service providers—in enterprise and open source versions.

Today, most organisations want to blend a variety of internal and external compute and storage resources to create an integrated hybrid cloud that allows workloads to be dynamically re-targeted to optimise for security, governance, compliance, price, performance and various service level characteristics. **END** 

References

- [1] Peter Mell, Timothy Grance, 'The NIST Definition of Cloud Computing' <http://csrc.nist.gov/publications/nistpubs/800-145/SP800-145.pdf>
- [2] Phil Wainewright, 'Private cloud discredited, part 2' <http://www.zdnet.com/blog/saas/private-cloud-discredited-part-2/1289>
- [3] Gartner, 'Private Cloud Computing: An Essential Overview'
- [4] David Cahill, Zynga Cloud Case Study: The journey to a real Private Cloud http://wikibon.org/wiki/Zynga_Cloud_Case_Study_The_Journey_to_a_real_private_cloud

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